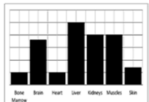
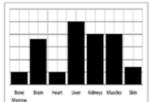


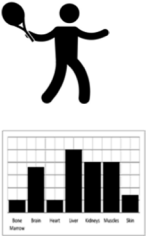
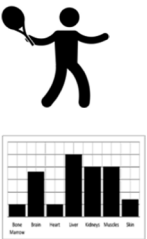
1.3.b. Linear motion, angular motion, fluid mechanics and projectile motion

Learners will develop their knowledge and understanding of linear motion by being able to define linear motion and be able to apply it as well as being able to calculate quantities of linear motion.

They will also be able to define angular motion and know about the creation of angular motion through the application of an eccentric force about one (or more) of the three axes of rotation. Again, learners will also be able to calculate angular motion and interpret graphs of angular velocity, moment of inertia and angular momentum.

Learners will develop their knowledge and understanding of fluid mechanics and the factors that impact the magnitude of air resistance (on land) or drag (in water) on a body or object. Projectile motion will also be understood with factors affecting the horizontal distance travelled by a projectile, as well as patterns of flight paths as a consequence of the relative size of air resistance and weight. Bernoulli's principle will be understood along with the application of projectile motion on the design of equipment and the use of spin in sport.

Topic Area	Content
*Linear motion  	• definition of linear motion. • creation of linear motion by the application of a direct force through the centre of mass • definitions, calculations and units of measurement for each of the following quantities of linear motion: ◦ distance ◦ displacement ◦ speed ◦ velocity ◦ acceleration/deceleration • plot and interpret graphs of linear motion: ◦ distance/time graphs ◦ speed/time graphs ◦ velocity/time graphs.
*Angular motion  	• definition of angular motion • creation of angular motion through the application of an eccentric force about one (or more) of the three axes of rotation: ◦ longitudinal ◦ frontal ◦ transverse • definitions, calculations and units of measurement for each quantity of angular motion: ◦ moment of inertia ◦ angular velocity ◦ angular momentum • factors affecting the size of the moment of inertia of a rotating body: ◦ mass of the body (or body part) ◦ distribution of the mass from the axis of rotation • the relationship between moment of inertia and angular velocity

Topic Area	Content
*Angular motion cont. 	- the conservation of angular momentum during flight in relation to the angular analogue of Newton's first law of motion - interpret graphs of angular velocity, moment of inertia and angular momentum.
*Fluid mechanics 	- factors that impact the magnitude of air resistance (on land) or drag (in water) on a body or object: - velocity - frontal cross-sectional area - streamlining and shape - surface characteristics.
*Projectile motion 	- factors affecting the horizontal distance travelled by a projectile: - height of release - speed of release - angle of release - free body diagrams showing the forces acting on a projectile once in flight: - weight - air resistance - resolution of forces acting on a projectile in flight using the parallelogram of forces - patterns of flight paths as a consequence of the relative size of air resistance and weight - parabolic (symmetrical) flight path - shot put - non-parabolic (asymmetric) flight path - badminton shuttle - The addition of lift to a projectile through the application of Bernoulli's principle: - angle of attack to create an upwards lift force on a projectile: - discus - javelin - ski jumper - design of equipment to create a downwards lift force: - F1 racing cars - track cycling - use of spin in sport to create a Magnus force, causing deviations to expected flight paths: - imparting spin to a projectile through the application of an eccentric force - types of spin: - top spin, side spin and back spin in tennis and table tennis - side spin in football - hook and slice in golf.